PRANJAL SINGH SESODIA

B.Tech. Mechanical Engineering, NSUT New Delhi (2024–2028) | R&D Intern, mPragati, IIT Delhi
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EDUCATION

Netaji Subhas University of Technology (NSUT), New Delhi — B.Tech., Mechanical Engineering **Aug 2024 – 2028**

- **CGPA: 8.07 / 10** (Semesters I–IV) | Department rank 19 / 198 (top 10%).
- **Completed:** Control Systems, Theory of Machines, Strength of Materials, Fluid Mechanics & Machines (O), Thermal Engineering I–II, Numerical Methods & Computation, Engineering Mechanics, Materials Science, Manufacturing Processes I–II, Metrology & SQC, Analog & Digital Electronics (O), Computer Programming (O). *O = 10/10, the highest grade on NSUT's scale.*
- **Current (Sem V):** Robotics, Machine Design, Product Design & Development, Advanced Manufacturing Processes, Industrial Engineering, Basics of Electric Vehicles.

Army Public School, Shankar Vihar, New Delhi — CBSE Class XII (PCM): 90% | Class X: 90% **2021 – 2023**

PUBLICATIONS & RESEARCH

- **Journal (first author, review):** Sesodia, P. S., & Dabass, V. (2026). *Robust Real-Time Levitation Gap Estimation for Hyperloop Pods.* Next Research (Elsevier). doi:10.1016/j.nexres.2026.102384. Reviews inductive, optical, capacitive, Hall-effect and UWB-radar gap sensing, with Kalman-filter (EKF/UKF) and sliding-mode observer methods, for EMS and EDS maglev under thermal drift and EMI.
- **Ongoing (independent):** Literature review on the thermal behaviour of robotic actuators and grippers. [Start date]
- **Poster, NSUT Research Conclave (2024–25):** "[Poster title]" — agricultural wheat-straw waste to biodegradable drinking straws; my first research project.
- **Provisional patent application (India, 2022):** EYEAR (see Projects).

RESEARCH & INDUSTRY EXPERIENCE

R&D Intern, mPragati, IIT Delhi — Supervisor: Prof. Dinesh Kalyanasundaram **Jun 2026 – Present**

- **6-DOF surgical-assistance arm & endoscope stabiliser:** building the prototype on open-loop digital servos and closing the position-feedback loop with vision (Python + OpenCV fingertip/instrument tracking); exploring stereo depth estimation; simulating in ROS 2 and MuJoCo. Next: mathematical modelling and policy training in MATLAB.
- **Patented elbow implant (pre-technology-transfer review):** modelled the implant and test jig as SolidWorks parts and assemblies and ran multi-body dynamics in ANSYS Mechanical; found a missed ulna-hinge trajectory flaw and am leading the kinematic redesign.
- **Prosthetic knee:** redesigned a jammed polycentric 4-bar knee linkage in SolidWorks (part & assembly) to remove its kinematic locking; validated with physical motion tests.

Research Intern, QuEST Lab, IIT Delhi **Dec 2025 – May 2026**

- Literature review on contactless, gamified assessment of physical and cognitive fatigue in clinicians, and on optical depth sensing.
- In a 4-member team, explored and developed a pump and lining system for arthroscopic fluid management.

Intern, IBM India — IBM Under-18 Internship (CBSE & Atal Innovation Mission) **Nov 2022**

- Invited by IBM to its national showcase at the Taj Hotel, New Delhi, where I presented EYEAR to the Union Minister of Education.

ENGINEERING PROJECTS

Tabletop Subsonic Wind Tunnel (6:1 contraction) — NSUT CPVSK project grant, ₹10,000 **Mar 2026 – Present**

- Did the design calculations, contraction CFD (ANSYS Fluent), structural FEA (ANSYS Static Structural) to cut wall thickness and printed mass, and the complete electronics. Design complete; optimisation and manufacturing under way.

Catalytic Converter, Team Daedalus Racing (BAJA SAEINDIA), NSUT — Junior Engineer **2025 – Present**

- Developed the team's catalytic converter and selected lower-cost materials to replace expensive catalyst elements and chemicals — a key factor in winning **1st place nationally** (₹30,000; 54 teams) at the mBAJA 2026 Sustainability Event.

EYEAR: AI-Enabled Hearing Assistant — Python, ESP32, OLED **2021 – 2022**

- Conceived and built a real-time speech-to-sign translation device. **Top 350 nationally**, ATL Marathon 2021–22 (Atal Innovation Mission); "Patent Pioneer" title from APS Shankar Vihar as one of 3 students whose provisional filings made it the first of 137 Army Public Schools to file. Finalist, Smart India Hackathon Junior (2022).

Quadruped Walking Robot — Change.Makers Bootcamp, IIT Delhi **May – Jun 2022**

- Selected among the top 25 school students nationally; led the 14-day build of a 4-bar-linkage quadruped (MIT Cheetah-inspired): mechanism assembly, servo actuation and gait programming.

LEADERSHIP & VOLUNTEERING

- **Mentor, Change.Makers Bootcamp, IIT Delhi (May–Jun 2025):** guided 8 students from problem definition to a working prototype for upcycling partially degraded Li-ion batteries in 3 weeks.
- **Team Daedalus Racing:** created the "Piston Cup Challenge" recruitment event; organised events at Moksha, Resonanz and Innovision (250+ participants each). **Member:** ASME NSUT, Enactus NSUT (2024–present).
- **President / Student Ambassador, Atal Tinkering Lab, APS Shankar Vihar (2021–22):** organised Tinkerfest, a 3-day hackathon with 50+ schools.

SKILLS

CAD / CAE: SolidWorks (parts & assemblies), ANSYS Mechanical (static structural, multi-body dynamics), ANSYS Fluent (CFD)
Robotics & simulation: ROS 2, MuJoCo, MATLAB **Programming & embedded:** Python (OpenCV), C, ESP32, Arduino
Fabrication: FDM & SLA 3D printing, OrcaSlicer, rapid prototyping **Languages:** English, Hindi